

P Douglas Yoder

PhD | Semiconductor Expert Witness and Litigation Consulting

Associate Professor, Georgia Institute of Technology

Independent Consultant, QuantumForge Semiconductor Modeling LLC

qfsm.contact@gmail.com | Atlanta, Georgia

quantumforgesemi.com | yoder.ece.gatech.edu

Professional Profile

Semiconductor physicist and computational device specialist with more than 30 years of experience in academic research, industrial R&D, and scientific consulting. Background combines research at Bell Laboratories and Agere Systems, development of commercial semiconductor simulation software, and research and teaching at Georgia Tech since 2003. Expertise centers on the physical operation, modeling, and performance of electronic and photonic semiconductor devices.

Technical Expertise

Device simulation and carrier transport: Technology computer-aided design (TCAD), full-band Monte Carlo methods, quantum transport, first-principles modeling, coupled electrothermal analysis, and numerical solution of semiconductor transport problems.

Photodetectors and avalanche devices: Avalanche photodiodes (APDs), single-photon avalanche diodes (SPADs), impact ionization, breakdown, gain and noise, ultraviolet detection, bandwidth, optical saturation, and photodiode nonlinearity.

III-nitride and wide bandgap semiconductors: GaN, AlGaN, InGaN, and AlN; heterostructures, polarization, high-field transport, self-heating, doping, and impurity-band physics.

Semiconductor photonics and emerging devices: Visible and ultraviolet lasers, vertical-cavity surface-emitting lasers, resonant tunneling structures, and millimeter-wave sources.

Technical Support for Counsel

Available for independent technical assessment, review of device documentation and scientific literature, analysis of simulation assumptions and performance claims, and explanation of semiconductor physics. Expert reports and testimony may be undertaken within the relevant technical expertise and agreed engagement scope.

Relevant questions include whether a proposed operating mechanism is physically consistent, whether a model supports its stated conclusion, and what device measurements establish about transport, gain, noise, bandwidth, or breakdown.

Education

PhD in Electrical Engineering, University of Illinois at Urbana-Champaign | 1993

Dissertation: First-Principles Simulation of Charge Transport in Semiconductors

MS in Electrical Engineering, University of Illinois at Urbana-Champaign | 1991

Thesis: New Approaches to Monte Carlo Simulation of Impact Ionization in Silicon

BSEE in Electrical Engineering, Cornell University | 1990

Distinction, highest honors

Professional Experience

Georgia Institute of Technology, Associate Professor | August 2003-present

School of Electrical and Computer Engineering. Research in computational electronics and photonics; development of physical models and simulation tools; teaching semiconductor device physics and supervision of graduate research.

QuantumForge Semiconductor Modeling LLC, Independent Consultant | Current practice

Semiconductor physics and modeling consulting, technical education, and evaluation of device concepts, simulation results, and physical performance limits.

Agere Systems, Member of Technical Staff | 2001-2003

Bell Laboratories, Member of Technical Staff | 1999-2001

Swiss Federal Institute of Technology, Member of Technical Staff | 1993-1999

Commercial Software and Scientific Consulting

Author of Degas, a full-band ensemble Monte Carlo semiconductor device simulator integrated into a commercial TCAD platform and distributed through Integrated Systems Engineering during 1995-2004. Scientific consultant to Integrated Systems Engineering AG, Zurich, 1994-1998; scientific consultant to Texas State University, Department of Materials Science, summer 2013.

Selected United States Patents

Doped-absorber graded transition enhanced multiplication avalanche photodetector. US 7,161,170 B1. Granted January 9, 2007. Inventor.

Epitaxial base layers for heterojunction bipolar transistors. US 8,809,912 B2. Granted August 19, 2014. Co-inventor.

Methods and systems for generating millimeter-wave oscillations. US 8,957,737 B2. Granted February 17, 2015. Co-inventor.

Professional Recognition and Teaching

Senior Member, IEEE (2005); U.S. Air Force Summer Faculty Fellowship (2009); ECE Teaching Fellow, Georgia Institute of Technology (2015). Technical instruction includes semiconductor devices and related electrical engineering subjects. German: fluent, spoken and written.

Selected Publications

Selected work relevant to semiconductor transport, numerical modeling, photodetectors, and photonic devices. A complete academic publication record is available through the links below.

Recent materials and device research

A. Billah and P. D. Yoder. "Impurity band formation in doped AlN: A Green's function approach." AIP Advances 16, 045003 (2026).

J. Sculley and P. D. Yoder. "Nucleation and Propagation of Traveling Charge Domains in a Planar AlGaIn/GaN Triode Structure." physica status solidi (a) 222, 2500094 (2025).

P. D. Yoder. "Theoretical Modeling of Avalanche Currents in GaN." In Gallium Nitride and Related Materials, Springer (2025).

K. Mehta and P. D. Yoder. "Philosophy of Approaching a Laser Design Problem: Illustrated by the Design of Ultraviolet Vertical-Cavity Laser Diodes." physica status solidi (a) 217, 2000154 (2020).

Photodetectors and avalanche devices

S. Choi et al., including P. D. Yoder. "Geiger-mode Operation of GaN Avalanche Photodiodes Grown on GaN Substrates." IEEE Photonics Technology Letters 21, 1526-1528 (2009).

H. Pan, A. Beling, H. Chen, J. C. Campbell, and P. D. Yoder. "A High-Linearity Modified Uni-Traveling Carrier Photodiode with Offset Effects of Nonlinear Capacitance." Journal of Lightwave Technology 27, 4435-4439 (2009).

P. D. Yoder. "Bandwidth and Charge Balancing of the Partially Depleted Absorber Photodiode." IEEE Journal of Quantum Electronics 43, 992-997 (2007).

P. D. Yoder and E. J. Flynn. "Linear theory of the quasi-unipolar photodiode." Journal of Lightwave Technology 24, 1937-1945 (2006).

Carrier transport and numerical methods

M. V. Fischetti, P. D. Yoder, M. M. Khatami, G. Gaddemane, and M. L. Van de Put. "Hot electrons in Si lose energy mostly to optical phonons: Truth or myth?" Applied Physics Letters 114, 222104 (2019).

P. D. Yoder, K. Gärtner, and W. Fichtner. "A generalized Ramo-Shockley theorem for classical to quantum transport at arbitrary frequencies." Journal of Applied Physics 79, 1951-1954 (1996).

Scientific Leadership

Topical Program Chair, International Workshop on Nitride Semiconductors (2024); Chair, International Conference on Superlattices, Nanostructures and Nanodevices (2014); Co-Chair, International Conference on Numerical Simulation of Optoelectronic Devices (2010).

[Full academic publication list](#) | [Google Scholar profile](#)

Academic contact: doug.yoder@gatech.edu | Office: +1 404 385 2652